

Raja Muhammad Bilal Arshad

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SUMMARY

Cybersecurity researcher and Purple Team analyst with hands-on experience across offensive testing, defensive monitoring, and applied security research. Leads and contributes to research in post-quantum cryptography, zero-knowledge systems, adversarial sensing, and AI-driven intrusion detection, alongside operating SIEM platforms and MITRE ATT&CK-based detection engineering in enterprise environments.

SKILLS

Defensive Security: SIEM (Wazuh, Splunk), IDS/IPS (Snort, Suricata), SOC Operations (L1/L2), Threat Detection, Incident Response, Firewall Engineering (pfSense, OPNsense), Log Analysis, Network Security Monitoring

Offensive Security: Network Penetration Testing, Web Security (OWASP Top 10), Privilege Escalation, Active Directory Exploitation, Wi-Fi Auditing, BloodHound, Reverse Engineering (Ghidra, x64dbg), MITRE ATT&CK

Applied Cryptography & Security Research: Post-Quantum Cryptography (PQC), PQC Benchmarking (NIST standards), Zero-Knowledge Proofs (zk-SNARKs, Groth16), Adversarial Machine Learning, Evolutionary Optimization (NSGA-II), Continual/Zero-Day Intrusion Detection

Tools & Technologies: Python, Bash, Nmap, Wireshark, Tcpdump, Zeek, Sysmon, Osquery, BloodHound, Burp Suite, Metasploit, Nessus, OpenVAS

PROFESSIONAL EXPERIENCE

Research Associate

Sep 2025 – Present

Cyberarians Research Lab, Punjab, Pakistan

- Conduct independent applied cybersecurity research spanning post-quantum cryptography, zero-knowledge systems, and adversarial security testing.
- Research zero-knowledge proof systems for migrating cryptographic state from classical ECDSA-based infrastructure to post-quantum architectures, including a Groth16 zk-SNARK circuit compiling 4.2M+ constraints.
- Investigate adversarial attacks against WiFi Channel State Information (CSI) sensing systems and design corresponding defense mechanisms.
- Prepare research findings for peer review at venues including IEEE TIFS and Elsevier BCRA.

Lead Researcher

Aug 2025 – Present

NeuraSec Research Group

- Lead a research group applying AI and evolutionary methods to cybersecurity, spanning continual intrusion detection and autonomous red teaming.
- Direct research on multi-objective evolutionary memory governance for continual zero-day network intrusion detection under concept drift.
- Direct development of ARTP, a safety-constrained autonomous red-team planner mapped to MITRE ATT&CK for automated attack-path generation.
- Coordinate experimental evaluation and benchmarking across research projects, guiding work toward peer-reviewed publication.

SOC Analyst

Jan 2026 – Apr 2026

ITSOLERA Pvt Ltd, Office 1786, Street 12, I-14/4 I-14, Islamabad, 46000, Pakistan

- Operated Wazuh SIEM for real-time log correlation, file integrity monitoring (FIM), and threat detection across enterprise systems.
- Configured and hardened pfSense and OPNsense firewalls, improving perimeter defense and secure VPN connectivity.
- Performed packet-level network analysis using Wireshark and Tcpdump to investigate anomalies and lateral movement.
- Monitored and responded to security alerts, supporting incident triage, investigation, and escalation workflows.
- Utilized tools such as Sysmon, Osquery, and Velociraptor for endpoint visibility and forensic investigation.

Cyber Security Engineer

Nov 2025 – Feb 2026

CybeCloud, Islamabad

- Conducted vulnerability assessments and penetration testing across network and web environments.

- Monitored security telemetry and investigated incidents, reconstructing attack timelines and recommending remediation.
- Tuned IDS/IPS rules and improved detection pipelines, reducing false positives and enhancing alert quality.
- Developed automation scripts to streamline repetitive security tasks and improve operational efficiency.

Campus Ambassador

Sep 2025 – Dec 2025

Hackviser Pvt Ltd, 71-75 Shelton Street, Covent Garden, London, United Kingdom(Remote)

- Delivered hands-on CTF training sessions simulating real-world cyber attacks.
- Conducted phishing awareness simulations to educate users on social engineering threats.

PROJECTS & RESEARCH

EvoGuard: Multi-Objective Evolutionary Memory Governance for Continual Zero-Day Network Intrusion Detection (IEEE SSCI 2027 – First Author – Accepted) 2027

- Proposed a memory-governance mechanism for continual NIDS, treating a fixed-size replay memory as a multi-objective portfolio rather than a single manually weighted score.
- Optimized retention across four objectives: current-distribution fidelity, historical threat retention, representation diversity, and drift-transition information.
- Applied NSGA-II with Pareto front construction and knee-point selection to govern sample retention under concept drift.

ARTP & AutoPENT-Bench (MCETS, Cyprus – First Author – Accepted) 2026 — GitHub

- Designed and implemented ARTP, a safety-constrained system converting multi-source reconnaissance (Nmap, BloodHound, web crawlers) into structured, multi-stage attack plans.
- Engineered a rule-based planning engine mapped to MITRE ATT&CK, generating automated kill-chains across Web, Active Directory, and Cloud environments.
- Developed a symbolic verification layer with constraint enforcement and human-in-the-loop gating to block unsafe or out-of-scope execution.
- Built AutoPENT-Bench, a reproducible benchmark with standardized metrics (coverage, path efficiency, stealth, safety) for evaluating autonomous pentesting agents.
- Modeled attack execution as a state-transition system with deterministic simulation for controlled evaluation of exploitation paths.
- Conducted large-scale evaluation (N=30 runs) across containerized environments (DVWA, Juice Shop, AD simulation, cloud emulation), analyzing success rates and failure modes.
- Identified key automation risks, including exploration drift, over-aggressive probing, and verifier dependency.
- Achieved 0.85 path efficiency in Active Directory attack graphs, identifying optimal privilege-escalation chains to domain dominance.

Zero-Knowledge State Migration for Post-Quantum Cryptography (BCRA Elsevier – Under Review – First Author) 2026 —

- Engineered a Zero-Knowledge proof system for migrating cryptographic state from classical ECDSA-based infrastructure to post-quantum architectures.
- Designed and implemented a Groth16 zk-SNARK migration circuit compiling over 4.2 million constraints, recording proof generation and verification performance.
- Benchmarked proof generation, verification latency, CPU utilization, and memory consumption to assess migration practicality at scale.
- Evaluated migration performance against NIST Post-Quantum Cryptography standardization requirements, demonstrating compatibility with emerging quantum-safe deployment models.
- Developed a reproducible framework for assessing the scalability, correctness, and overhead of zero-knowledge-based cryptographic transitions.

Adversarial WiFi CSI Sensing & MeshGuard (IEEE TIFS – Under Review – First Author) 2026

- Studied denial-of-service and data-poisoning attacks against WiFi Channel State Information (CSI) sensing using a six-node ESP32-S3 mesh testbed.
- Designed MeshGuard, a defense mechanism countering the identified attack vectors.
- Verified benchmarks across a rigorous experimental workflow to evaluate attack impact and defense effectiveness.

Simulating a Full-Scale Cyber Warfare in a Virtualized Datacenter 2025 — GitHub

- Designed a virtualized cyber range using Proxmox Servers, pfSense, Suricata, and Splunk SIEM.
- Implemented VLAN segmentation and simulated attack-defense scenarios.
- Built automated alerting and response pipelines for SOC operations.

- Engineered a custom ESP32-S3 offensive security device emulating Flipper Zero capabilities for wireless and hardware-level security testing.
- Developed Wi-Fi auditing modules: deauthentication attacks, WPA/WPA2 handshake capture, rogue AP deployment, and real-time reconnaissance.
- Implemented BLE and RF scanning modules for detecting and analyzing nearby wireless devices.
- Integrated credential harvesting and packet sniffing using low-level network interfaces and custom firmware.
- Designed a modular C/C++ firmware architecture, extensible to additional attack vectors and payloads.
- Built an OLED interface for real-time attack control, logging, and payload selection.
- Used SD card storage for payload management, data logging, and offline analysis.
- Simulated red team scenarios including Evil Twin attacks, MITM setups, and wireless intrusion testing.
- Optimized power consumption and hardware efficiency for portable field deployment.

CERTIFICATIONS

- Certified in Cybersecurity (CC) – ISC2
- Certified Cybersecurity Educator Professional (CCEP)
- Certified Threat Intelligence & Governance Analyst (CTIGA)
- Palo Alto Network Cybersecurity
- Google IT Support
- Certified Penetration Tester Specialist (CPTS) (In Progress)
- Certified Associate Penetration Tester (CAPT) (In Progress)

ACHIEVEMENTS

- FAST Intra-University Project Exhibition 2025 – 1st Position
- FAST Intra-University Project Exhibition 2026 – 1st Position
- National Idea Cup (NCCS) 2025 – 3rd Position
- DAIRA National CTF 2025 – 3rd Position
- Innovate Pakistan 2026 – Project Shortlisted
- DAMUN 2025 (UNGA Committee, DAIRA) – 1st Position (Best Delegate)
- AMUN 2025 (UNSC Committee) – 2nd Position (Outstanding Delegate)
- NUMUN 2025 (UNSC Committee, FAST Lahore) – 2nd Position (Outstanding Delegate)
- AMUN 2026 (UNSC Committee) – 1st Position (Best Delegate)
- FMUNC 2026 (UNSC Committee) – 1st Position (Best Delegate)

EDUCATION

FAST National University (NUCES)

BS Computer Science